

## ***Experiment 8: A Algorithm\****

### **Aim**

To implement the A\* Search Algorithm using Python.

### **Objective**

To find the shortest path between two nodes.

### **Algorithm**

1. Select the start node.
2. Calculate  $f(n)=g(n)+h(n)$ .
3. Expand the node with the lowest cost.
4. Continue until the goal is reached.
5. Display the shortest path.

### **Flowchart**

Start

|

Start Node

|

Calculate Cost

|

Goal?

/ \

No Yes

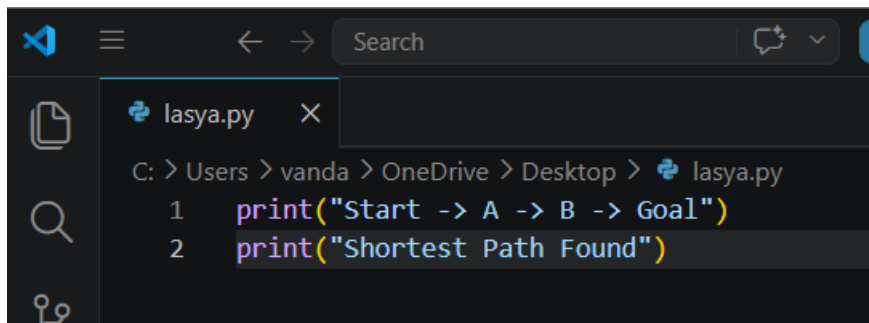
| |

Expand Stop

|

Repeat

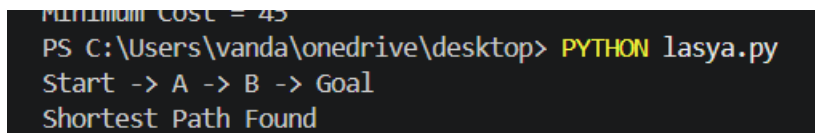
### **Code**



The screenshot shows a code editor with a dark theme. The file explorer on the left shows a file named 'lasya.py'. The main editor area shows the following code:

```
C: > Users > vanda > OneDrive > Desktop > lasya.py
1 print("Start -> A -> B -> Goal")
2 print("Shortest Path Found")
```

Output:



The screenshot shows a terminal window with the following output:

```
Minimum Cost = 45
PS C:\Users\vanda\onedrive\desktop> PYTHON lasya.py
Start -> A -> B -> Goal
Shortest Path Found
```

## Result

The shortest path was found successfully.

## Conclusion

A\* efficiently finds the optimal path using cost and heuristic values.